

Dark Horse Prototype Plan

Aryan Sood, Himansu Pathak, Junyu Zhou, Ilham, Jolie,
Gavin Tey, Lim Soong En

Reimagining a Luxury Hotel Check-In Experience

Background

- Large luxury hotels manage high guest volumes, especially during peak arrivals
- Guests require immediate assistance upon arrival:
 - Wayfinding to rooms and facilities
 - Luggage support
 - Basic hotel information
- Front-of-house staff handle simultaneous demands, limiting personalised attention
- Waiting and navigation issues introduce early friction in the guest journey

First impressions are formed before guests reach their room.



[Belagio Hotel Lobby](#)

User Needs & Rationale

- **Seamless Luggage Handling:** Guests want their bags taken care of immediately upon arrival without physical strain
- **Effortless Relaxation:** Guests want to feel pampered and "do nothing" while moving through the hotel
- **Instant Information:** Guests want their questions answered in real-time without searching for staff
- **Intuitive Navigation:** Guests need a clear, stress-free way to find their room and hotel amenities

Personal Robot Assistant

Functionalities:

- **Personal assistant in the form of your favourite character**
- **AI Real-Time Q&A**
- **Autonomous Luggage Robot which could be called by the AI assistant**
- **Tour guiding and Wayfinding capabilities**



The Core Idea (WHAT)

The Experience:

1. Guest arrives; a small, friendly drone (The "AeroHost") flies and descends to greet them.
2. Facial recognition confirms identity instantly.
3. The robot says, "Welcome, [Name], follow me.", and asks if the guest needs help with their luggage
 - i. If so, it calls a buddy ground robot that is able to roll the luggage around
4. It physically leads them through the lobby, to the elevator, and to their room.
5. The robot is able to answer any questions.

Prototyping Strategy (HOW)

The Rig:

- **Visual:** A foam/cardboard drone model attached to a long, pole. Holographic visual is above the drone.
- **Movement:** A team member (dressed in black or standing behind) holds the pole to simulate hovering, bobbing, and leading.
- **Audio / AI:** A phone inside the "drone" model running a custom GPT / model to converse with the guest based on sample hotel policies and details.



Tests:

- Analyse the user's feedback and initial response to a small flying robot at eye level
- Analyse how the conversations flow when both the robot and the guest are moving

Assumptions to test

- **The "Creepiness" Threshold:** We assume guests will feel *cared for* rather than *surveilled* or *threatened*
- **Safety/Proximity:** We assume guests will trust the robot enough to walk close to it without flinching (the "fear of crashing" factor).
- **Intuitive Communication:** We assume guests will naturally talk to a flying object for questions ("Where is the pool?") without needing a screen interface for interaction.
- People want to be engaged throughout their entire check-in process rather than being alone

Stanford and NUS Team Roles

To execute this rapidly, we will divide the physical and digital workloads:

- **Team A:** Aryan, Jolie, Gavin, Himansu
 - Building the rig for the drone and the physical structures
- **Team B:** Junyu, Soong En, Ilham
 - The AI functionalities of the robot and conversational responses

Background

GUEST NEEDS

Our

Immersive Check-in Museum Experience



Background

- Hotels with large volumes
- Priority to guest satisfaction especially so for luxury hotels
- According to Cornell's Centre for Hospitality. Research, guest satisfaction reduces by 47% the moment the check in takes longer than 5 minutes
- Want to reduce waiting time



Belagio Hotel Lobby

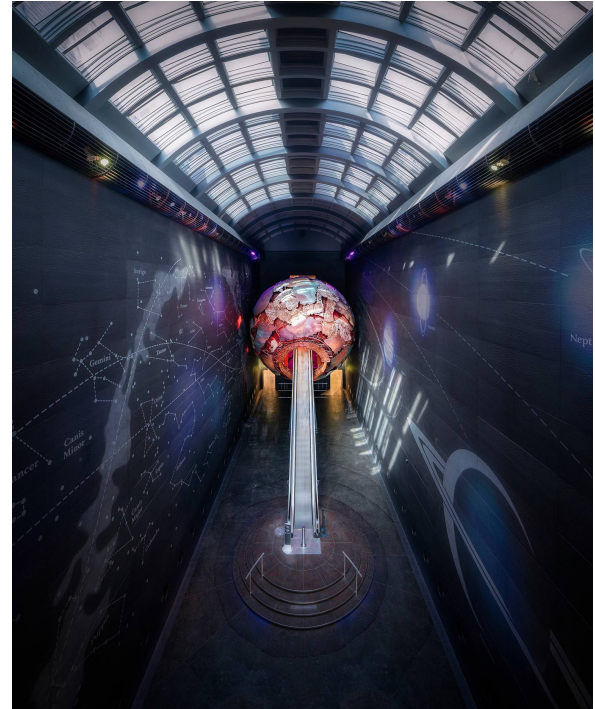
Our Core Idea

We propose supplementing the traditional front desk. We will simulate a "travelator" experience where the journey from the hotel entrance to the elevator becomes an immersive, museum-grade exhibition.



The Core Idea (WHAT)

- **The Mechanism:** Guests travel through a corridor (simulated using trolleys) where walls act as dynamic displays.
- **The Content:** Projectors visualize the hotel's history, service & amenities, heritage, local tourist spots, and/or sustainability efforts. An AI narrator would curate the voice over experience and answer any queries.
- **The Transaction:** Biometric scans (face/passport) occur invisibly during this journey. Interest is detected via gesture tracking, allowing the system to upsell amenities or customize the room before the guest even arrives at their door.



[Explore](#)

Rationale & User Needs (WHY)

This concept directly addresses the tension between efficiency and experience in the luxury sector.

- **Need for Seamlessness:** High-end guests view administrative pauses as friction. This prototype tests if we can make check-in invisible.
- **Need for Narrative Connection:** Guests often feel disconnected from the local culture inside a generic luxury lobby. This turns the arrival into an immediate cultural "download."
- **Business Viability:** By using gesture detection to track interest (e.g., a guest points at a spa visual), we open a new, non-intrusive channel for upselling services.



[r/vegas: Caesar's Palace](https://www.reddit.com/r/vegas/)

Prototyping Strategy (HOW)

We will construct a low-fidelity physical experience to test the *feeling* of the journey rather than the final technology.

- **Physical Setup:** We will simulate the "travellator" using a rolling cart or trolley system to move the user through a darkened hallway or enclosed space.
- **Visuals:** We will use standard projectors to "flash" images & videos on the walls, as well as a small-scale hologram.
- **Wizard of Oz Testing:** We will not build real AI gesture recognition yet. A team member will act as the computer, observing the user and logging "purchases" or "check-in completion" based on the user's physical movements.
- **Metric for Success:** We will measure "delight" versus "confusion." Do users feel taken care of, or do they feel out of control without a front desk agent?



[Alamy](#)

Assumptions

- The Immersive Entry is an exclusive "**Fast Track**" accessible only via the Hotel's app.
- By shifting data entry to *before* arrival, we remove 100% of the administrative friction from the physical lobby, allowing the "Museum" to focus purely on narrative and emotion
- This ensures that the room key dispensed at the end is accurate to the guest's room and check-in details.

Stanford and NUS Team Roles

To execute this rapidly, we will divide the physical and digital workloads:

- **Team A:** Aryan, Jolie, Gavin
 - Construction of the physical rig (trolley/hallway setup).
 - Sourcing projectors and managing the physical "Wizard of Oz" testing with users.
- **Team B:** Junyu, Soong En, Ilham
 - **Content Creation:** Curating the "museum" assets (historical photos, sustainability facts) and stitching them into a cohesive video loop for projection.
 - **UI/UX Mockup:** Designing the "pre-arrival" mobile interface where users might upload their passport photo to enable the facial recognition feature.